

Introduction to Big Data Analytics

Chapter 3

Need of Big Data

- ▶ Following are key terms and their meanings, which are essential to understand the topics discussed in this chapter:
- ❑ Application
- ❑ Application Programming Interface (API)
- ❑ Data Model
- ❑ Data Repository
- ❑ Data Store
- ❑ Distributed Data Store
- ❑ Database
- ❑ Table
- ❑ Flat File
- ❑ CSV File
- ❑ Name Value Pair
- ❑ Key Value Pair
- ❑ Hash Key Value Pair
- ❑ Stream Analytics

- ❑ Database Management System (DBMS)
- ❑ Relational Database
- ❑ Data Warehouse
- ❑ Data Mart
- ❑ Batch Processing
- ❑ Streaming transaction Processing
- ❑ Service -Oriented Architecture (SOA)
- ❑ Descriptive Analytics
- ❑ Predictive Analytics
- ❑ Prescriptive Analytics
- ❑ Cognitive Analytics

Big Data Examples

- ▶ **Black Box Data:** It is a component of helicopter, airplanes, and jets, etc. It captures voices of the flight crew, recordings of microphones and earphones, and the performance information of the aircraft.
- ▶ **Social Media Data:** such as Facebook and Twitter hold information and the views posted by millions of people across the globe.
- ▶ **Stock Exchange Data:** holds information about the 'buy' and 'sell' decisions made on a share of different companies made by the customers.
- ▶ **Power Grid Data:** holds information consumed by a particular node with respect to a base station.
- ▶ **Transport Data:** includes model, capacity, distance and availability of a vehicle.
- ▶ **Search Engine Data:** Search engines retrieve lots of data from different databases.

Who is a Big Data developer

- ▶ A Big Data Developer is the one who is responsible for developing Hadoop applications (or others). It serves the big data needs of an organization he is working in, and works to solve the big data problems and requirements.

Roles and Responsibilities of Big Data Developer

Loading the data from disparate data sets.

High-speed querying.

Propose best practices and standards.

As a big data developer, your role is to design, build, install, and support Hadoop (or Others..).

He maintains security and data privacy.

Big Data developer manages and deploys HBase.

He performs an analysis of the vast number of data stores and uncovers insights.

Big Data Developer is responsible for Hadoop development and implementation.

He is responsible for creating scalable and high-performance web services for tracking data.

Big Data developer translates complex technical and functional requirements into detailed designs.

He proposes design changes and suggestions to various processes and products.

Skills Required to Become Big Data Developer

- 01 *Big Data Frameworks*
- 02 *Real-time Processing Framework*
- 03 *SQL based Technologies*
- 04 *NoSQL based Technologies*
- 05 *Programming Languages*

Designing Data Architecture

- Big Data Architecture is the logical layout/structure of how Big Data will be stored, accessed, and managed within IT environment.
- Architecture defines how Big Data solution will work, the core components (**Hardware, database, software, storage**) used, **flow of information**, and **security**.
- Big Data Architecture design task simplifies when using the logical layers approach.
- The Following figure shows **the logical layers and the functions** which are considered in Big Data architecture.
 - The five vertically aligned boxes on the left show the layers
 - Horizontal boxes show the functions in each layer
- Data Processing architecture consists of five layers:
 1. Identification of data sources
 2. Acquisition, ingestion, extraction, pre-processing, transformation of data
 3. Data storage at files, servers, cluster, or cloud
 4. Data processing
 5. Data consumption

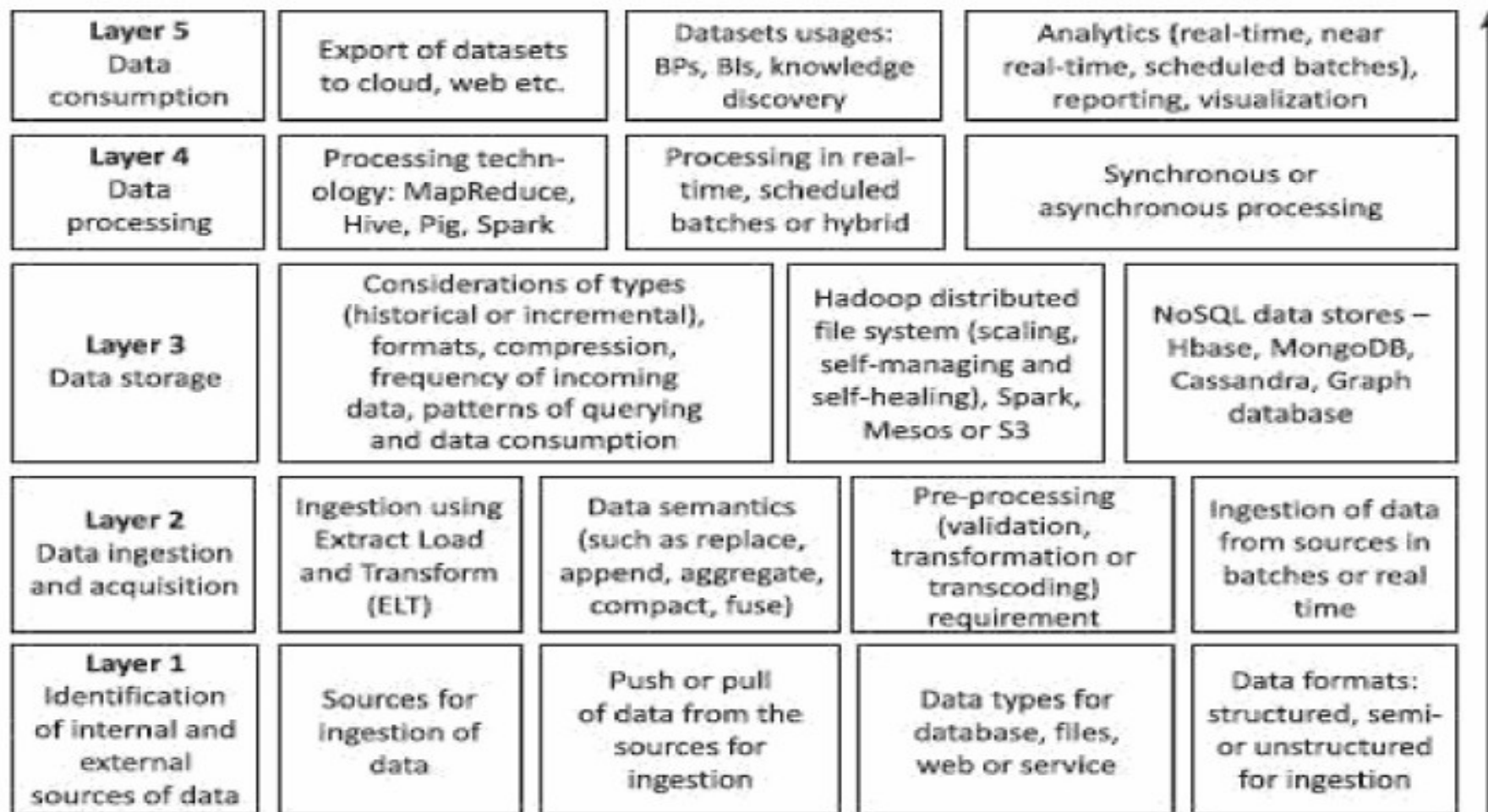
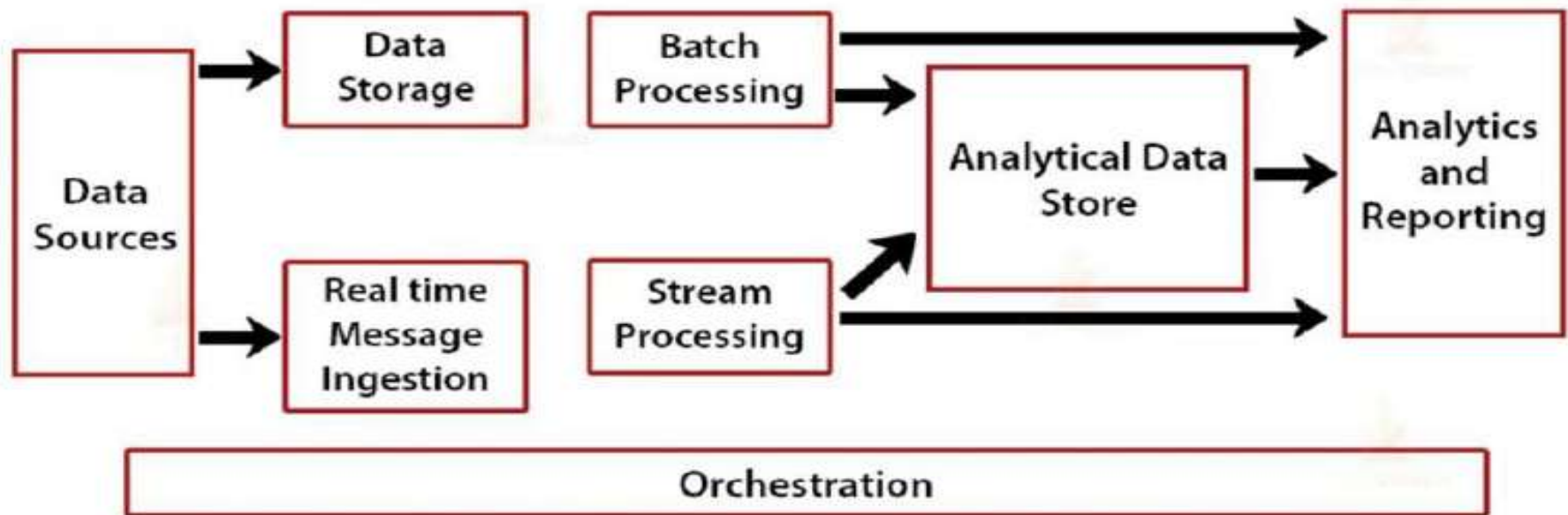


Figure 1.2 Design of logical layers in a data processing architecture, and functions in the layers

- Data consumed for applications such as business intelligence, data mining, discovering patterns/clusters, Artificial intelligence (AI), machine learning (ML), Text Analytics, descriptive and predictive analytics and data visualization.
- **Logical layer 1 (L1)** is for identifying data sources which are external, internal, or both.
- Layer 2 (L2) is for data ingestion. Data ingestion means a process of absorbing information, is the process of obtaining and importing data for immediate use or transfer. Ingestion may be in batches or in real time using pre-processing or semantics.
- The L3 layer is for storage of data from the L2 layer.
- The L4 layer is for data processing using software, such as Mapreduce, Hive, Pig or Spark.
- The top layer L5 is for data consumption.
- Data is used in analytics, visualizations, reporting, , export, to cloud or web servers.

- L1 considers the following aspects in a design:
 - Amount of data needed at ingestion layer 2 (L2)
 - Source data-type: Database, files, web or service
 - Source formats: : semi-structured, unstructured, or structured.
- L2 considers the following aspects:
 - Store and use the data as generated, or in batches. Batch processing is using discrete datasets at scheduled or periodic intervals of time.
- L3 considers the following aspects:
 - Data storage type (historical or incremental), format, compression, incoming data frequency, querying patterns and consumption requirements for L4 or L5.
 - Data storage using Hadoop distributed file system or NoSQL data stores - Hbase, Cassandra, MongoDB.
- L4 considers the following aspects:
 - Data processing software such as MapReduce, Hive, Pig, Spark.
 - Processing in scheduled batches or real time or hybrid.
 - Processing as per synchronous or asynchronous processing requirements at L5.
- L5 considers the following aspects:
 - Datasets usages for reporting and visualization,
 - Analytics (real time, batches)
 - Export of datasets to cloud, web, or other systems.

Architecture of Big Data



Managing Data for Analysis

- Data managing means enabling, controlling, protecting, delivering and enhancing the value of data and information.
- Data management functions include:
 1. Data assets creation, maintenance, and protection.
 2. Data governance which includes establishing the processes for ensuring the availability, usability, integrity, security and high-quality of data.
 3. Data architecture creation, modelling and analysis
 4. Database maintenance, administration, and management system.
 5. Managing data security, data access control, Privacy and security
 6. Managing the data quality
 7. Managing documents, records, and contents
 8. Integrated data management
 9. Data warehouse management
 10. Data mining algorithms

Data Sources, Quality, Pre-Processing and Storing

1. Data Sources:

Sources can be external, such as sensors, trackers, web logs.

Sources can be machines, which source data from data-creating programs.

Data sources can be structured, semi-structured, multi-structured or unstructured.

Sources can internal, can be data repositories, such as database, relational database, flat file, spreadsheet, web server.

Structured data sources:

Data source can be a file, database, or streaming data.

The source may be on the same computer or a networked computer.

Examples: SQL server, MySQL, Microsoft access database, Oracle DBMS, Amazon simple DB.

Unstructured data sources:

Are distributed over networks, the sources are of file types such as .txt (text file), .csv (comma separated values file)

Sensors, Signals, and GPS:

Sources can be sensors, signal from machines, devices, or controllers.

2. Data Quality:

High quality means data with five R's: Relevant, rich, reconcilable, robust, and reliable.

A reference can be made to a set of values of quantitative or qualitative conditions, which must be specified to say that data quality is high or low.

Data integrity:

Refers to the maintenance of **consistency** and **accuracy** in data over its usable life.

Software which store, process, or retrieve the data should maintain the integrity of data

Data Noise, Outliers, Missing, and duplicate values :

Noisy data means data having large additional information, Result of data analysis is affected due to noisy data.

A statistical analysis of deviation can select the noise in data and true values can be retrieved.

Outliers:

An *outlier* in data refers to data which appears to not belong to the dataset, data that is outside an expected range.

Actual outliers need to be removed from the dataset.

The outliers are a result of human data-entry errors, programming bugs.

Missing Values:

Missing value implies data *not appearing* in the data set.

Duplicate Values:

Duplicate value implies the same data *appearing two or more times* in the dataset.

3. Data Pre-Processing:

Data preprocessing is an important step at the ingestion layer.

Data preprocessing is a must before data mining and analytics, and before running a Machine Learning (ML) Algorithm.

Pre-processing needs are:

1. Dropping out of range, inconsistent and outlier values.
2. Filtering unreliable, irrelevant and redundant information.
3. Data cleaning, editing and reduction.
4. Data validation, transformation or transcoding.
5. ELT (Extract, Load, Transform) processing.

Data Cleaning :

Refers to the process of removing, or correcting incomplete, incorrect, inaccurate, irrelevant parts of the data.

Data cleaning Tools:

Data cleaning is done before mining of data. Incomplete or irrelevant data may result into misleading decisions. Data can generate in a system in many formats when it is obtained from the web. Data cleaning tools help in refining and structuring data into usable data. Examples of such tools: **OpenRefine and DataCleaner**

Data Editing:

- ▶ Data *Editing* refers to the process of reviewing and adjusting the acquired datasets. The **editing controls the data quality**.
- ▶ Editing methods are: interactive, selective, automatic, aggregating, distribution.

Data Reduction:

- ▶ Data *reduction* enables the transformation of acquired information into an ordered, correct, and simplified form.
- ▶ The basic concepts is the reduction of the diverse amount of data, and **use of the meaningful parts**.
- ▶ The reduction uses editing, scaling, coding, sorting ,collating, smoothening, interpolating, and preparing tabular summaries.

Data Wrangling:

- ▶ Refers to the process transforming and mapping the data. Mapping enables data into another format which makes it valuable for analytics and data visualizations.

Data format used during pre-processing:

Examples of formats for data transfer from a) data storage b) analytics application c) services d) cloud can be:

- Comma separated vlaues (csv)
- Java script object notation (JSON) as batches of object arrays or resource arrays
- Tag length value (TLV)
- Key-value pairs
- Hash-key-value pairs

Consider the example of a table in a grade sheet. A CSV file is easily understandable when the table's first row specifies the column heads. Three columns of the first row are Subject Code, Subject Name and Grade and three columns of the second row are CS101, "Theory of Computations" and 7.8, as shown below:

Subject Code	Subject Name	Grade
CS101	"Theory of Computations"	7.8
CS102	"Computer Architecture"	7.2
-	-	-

SOLUTION

The first and second lines in the CSV file are:

Subject Code, Subject Name, Grade

CS101, ""Theory of Computations"", 7.8.

CS102, ""Computer Architecture"", 7.8.

The two consecutive double-quotes mean that one of the double quotes is retained in the text "Theory of Computations". That one specifies that characters are inside the double quotes and represent a string.

4. Data Store Export to Cloud:

□ Data Storage and Management: Traditional systems (Structured or Semi-Structured Data)

Traditional systems use structured or semi-structured data, the following example explains the sources and data store of structured data.

What are the sources of structured data store?

SOLUTION

The sources of structured data store are:

- Traditional relational database-management system (RDBMS) data, such as MySQL DB2, enterprise server and data warehouse
- Business process data which stores business events, such as registering a customer, taking an order, generating an invoice, and managing products in pre-defined formats. The data falls in the category of highly structured data. The data consists of transaction records, tables, relationships and metadata that build the information about the business data.
- Commercial transactions
- Banking/stock records
- E-commerce transactions data.

- ❑ The following example explains the sources and data store of semi-structured data.

Give examples of sources of data store of semi-structured data.

SOLUTION

Examples of semi-structured data are:

- XML and JSON semi-structured documents^{7,8}
- A comma-separated values (CSV) file. The CSV stores tabular data in plain text. Each line is a data record. A record can have several fields, each field separated by a comma. Structured data, such as database include multiple relations but CSV does not consider the relations in a single CSV file. CSV cannot represent object-oriented databases or hierarchical data records. A CSV file is as follows:

```
Preeti,1995,MCA,Object Oriented Programming,8.75
```

```
Kirti,2010, M.Tech., Mobile Operating System, 8.5
```

Data represent the data records for columns and rows of a table. Each row has names, year of passing, degree name, course name and grade point out of 10. Rows are separated by a new line and the columns by a comma.

JSON Object Data Formats: CSV does not represent object-oriented records, databases or hierarchical data records. JSON and XML represent semi-structured data and represent object-oriented and hierarchical data records. Example 3.5 explains CSV and JSON objects and the hierarchical data records in the JSON file format.

SQL (Structured Query Language)

- ▶ A RDBMS uses SQL
- ▶ SQL is a language for viewing or changing (update, insert, or delete) databases.
- ▶ It is a language for data access control, schema creation and data modifications.
- ▶ SQL can embed within other languages using SQL modules and libraries.

Big Data Technologies

- ▶ There are various technologies in the market from different vendors including Amazon, IBM, Microsoft, Huawei, etc., to handle big data.
- ▶ There are the following two classes of technology:
 - Operational Big Data (NoSQL, MongoDB, Cassandra, HBas)
 - Analytical Big Data (Hadoop, Spark, Data Mining, MachineLearning)



Big Data Technologies:

These two classes of technology are complementary and frequently deployed together

Operational Big Data:

- These include systems like NOSQL as MongoDB that provide operational capabilities for real-time, interactive workloads where data is primarily captured and stored.

Analytical Big Data :

- These includes systems like Massively Parallel Processing (MPP) database systems and MapReduce that provide analytical capabilities for complex analysis that may touch most or all of the data.

Big Data Handling Techniques

- ▶ Following are the techniques deployed for big data storage, applications, data management, and mining and analytics:
 1. Huge data volumes storage, data distribution, high speed networks, and high performance computing
 2. Applications using open source, reliable, scalable, distributed file system, distributed database, parallel and distributed computing systems such as Hadoop or Spark.
 3. Open source tools which are scalable and provide virtualized environment, clusters of data nodes, task and thread management.
 4. Data management using NoSQL, document database, column-oriented database, graph database, and other form of databases.
 5. Data mining and analytics, data retrieval, data reporting, data visualization, and machine-learning big data tools

Question

What do you mean by the data noise, outliers, duplicate data and data anomaly? Why does the filtering require during pre-processing? Explain the following data processing steps: inspecting, cleaning, transforming, modeling and visualizing data.

<https://techvidvan.com/tutorials/big-data-applications/>

Big Data Applications in Different Sectors

